

From Zero to First Job

No fluff. Just what works.

The Reality Check

- ✗ You don't need 20 tools
- ✗ You don't need a Master's degree
- ✓ You need: SQL + Python + One Cloud
- ✓ You need: 2-3 strong projects

Must-Master Concepts

(In this order)

- ## 1. SQL (Window functions, CTEs, query optimization)

- ## 2. Python/PySpark (intermediate level)

- ### 3. Databricks/Snowflake

- #### 4. Data Modeling (Star Schema, SCD Type 2)

- ## 5. ETL/ELT pipelines (Airflow, dbt)

Must-Master Concepts

(Continued)

6. Cloud (AWS or GCP or Azure - pick ONE)

7. Kafka basics (producer/consumer)

8. Spark internals (partitioning, shuffling)

9. Data Quality (Great Expectations, dbt tests)

10. Git + CI/CD for data pipelines

Python Essentials

Learn only what you need

Pandas (read, filter, transform, write)

File handling (CSV, JSON, Parquet)



APIs (requests library)



Error handling (try/except)



Functions + Classes (basics)

Keep it simple and hands-on

Tools You Must Know

Pick ONE from each row

Cloud

AWS or GCP or Azure

Compute

Spark or Snowflake

Orchestration

Airflow or Prefect

Storage

S3 or ADLS or GCS

Streaming

Kafka or Kinesis

Don't overthink. Just pick and learn.



API → S3 → Spark → Redshift/BigQuery

Python

Airflow

SQL

Kafka → Spark Streaming → Dashboard

Kafka

PySpark

Streamlit

Automated data validation + alerts

Great Expectations

Airflow

Resume One-Liners

Copy these formats

- Built end-to-end ETL pipelines using Spark and Python
- Optimized pipelines that cut runtime by 30%
- Designed data models used in production dashboards
- Automated ingestion from 3+ data sources
- Reduced query cost by 60% using partitioning
- Implemented SCD Type 2 for 2TB customer table

Interview Questions

What they actually ask

1. Design a SCD Type 2 table
2. Optimize a 10-min query to <10 seconds
3. Handle 100GB skewed data in Spark
4. Build API → Bronze → Silver → Gold pipeline
5. Explain partitioning vs bucketing vs Z-ordering
6. "Your Airflow DAG failed at 3 AM - debug it"

- ✓ Build small, daily

- ✓ Reach out for referrals (DM people!)

✗ Don't wait to feel "ready"

Start today. Not tomorrow.

Learn from the Best

Ankit Bansal (YouTube)

SQL interview questions & patterns

Darshil Parmar (YouTube)

Real project tutorials

Zach Wilson (LinkedIn)

DE career & Bootcamps

Ansh Lamba (Youtube)

DE projects & Interview prep

Sumit Mittal (Youtube)

Interview prep & Spark

Shashank Mishra (LinkedIn)

Big Data and DE

Krish Naik (YouTube)

Python & ML pipelines